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van der Waals Volumes and Radii

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Intermolecular van der Waals radii of the nonmetallic elements have been assembled into a list of "recommended" values for volume calculations. These values have been arrived at by selecting from the most reliable X-ray diffraction data those which could be reconciled with crystal density at 0°K. (to give reasonable packing density), gas kinetic collision cross section, critical density, and liquid state properties. A qualitative understanding of the nature of van der Waals radii is provided by correlation with the de Broglie wave length of the outermost valence electron. Tentative values for the van der Waals radii of metallic elements—in metal organic compounds—are proposed. The paper concludes with a list of increments for the volume of molecules impenetrable to thermal collision, the so-called van der Waals volume, and of the corresponding increments in area per molecule.

Table of Nomenclature

- A_w Surface area of molecules (based on model of Fig. 3) per mole (cm.²/mole)
- b Covalent bond radius, Å.
- C Normalization constant in eq. 1
- d_w Nonbonded internuclear distance between atoms of neighboring molecules
- $E°$ Standard energy of vaporization, defined in ref. 2
- h Planck constant
- I_0 First ionization potential
- l Bond distance, Å.
- m_e Rest mass of an electron
- N_A Avogadro number
- r Distance from atom nucleus
- r_0 Distance between like nonbonded atoms (molecules) at potential energy minimum
- r_b van der Waals radius estimated by Pauling's approximation, $r_b = b$ + const.
- r_w' van der Waals radius derived from nonbonded contact distance in crystals
- r_w van der Waals radius
- $\bar{r}_w$ Mean van der Waals radius for volume calculations involving anisometric atoms
- V_c Molal volume
- V_0 Molal volume at 0°K.
- V_c Molal volume at critical temperature and pressure
- V_w van der Waals volume (calculated on the basis of Fig. 3)
- Z Number of nearest neighbors of a molecule
- $\delta(\ )$ Decrement of V_w or A_w (as indicated) due to intramolecular crowding or hydrogen bonding
- $\lambda_B \equiv h\sqrt{m_e I_0}$ = de Broglie wave length of outermost valence electron
- $\rho_0^* \equiv V_w/V_0$ = packing density at 0°K.
- $\rho_c^* \equiv V_w/V_c$ = packing density at critical point
- σ Distance between like atoms (molecules) at steepest ascent of repulsion branch of potential energy well (as obtained from gas properties by means of Lennard-Jones and Devonshire theory)
- ψ^2 Charge density (probability of finding an electron) at distance r from the atom nucleus

Purpose and Scope

The primary purpose of the present investigation is the calculation of the volume occupied by a molecule, *i.e.*, impenetrable for other molecules with thermal energies at ordinary temperatures. This volume, called here the van der Waals volume (V_w), is to serve

as reducing parameter in the study of the physical properties of condensed phases.[1,2] The calculation of V_w assumes a knowledge of bond distances, bond angles, and the contact distances [*i.e.*, intermolecular van der Waals radii (r_w)] and shapes characteristic of atoms in various molecular configurations. While most of the important bond distances and angles in organic molecules are reasonably well known and recorded,[3,4] only a few and semiquantitative contact (= nonbonded intermolecular) distances in crystals have been collected.

The first task at hand was therefore the collection of contact distances from reliable X-ray diffraction data in the literature. Selection of the "best" values from the resulting mass of data is of necessity an arbitrary procedure, which will be discussed in the body of this paper. Two important approximations have been made which can only be excused by the desire to obtain results now; namely, the contact distances have not been corrected[5] to 0°K. and all atoms have been treated as spheres and sphere segments, although it is well known that many are more nearly pear-shaped. The reason for the first approximation is the absence of data regarding the often anisotropic thermal expansion coefficients of the crystals from which the contact distance data were obtained. Spherical shapes have been assumed because of the absence of generally agreed pear shapes for the various atoms and the mathematical complexity of testing various alternative shapes for internal consistency. The values of V_w presented in this paper are therefore subject to further improvement. However, they have proven sufficiently useful, even in their present state, so that their publication appears justified.

General Principles

Assumption of the existence of a defined spatial extent of atoms is common to kinetic gas theory and X-ray crystallography. The extent of agreement between the dimensions of the rare gas atoms produced by both approaches can be considered as a measure of the status of kinetic gas theory.[6] However, this test is restricted to the rare gases because the crystals of metals, the only other monatomic species, are held together by covalent bonds and therefore do not permit comparison of the interatomic distances during thermal collisions of gas atoms with those prevailing in the solid.

This observation brings us to one of the tacit assumptions of this inquiry, the invariance of the van der Waals radius of an atom under the most drastic environmental changes, *i.e.*, irrespective of its chemical combination and of its nearest nonbonded neighbors as well as of the phase state in which it is found. Closer examination shows that this assumption is surprisingly valid for heavy atoms, but is not very good with atoms containing only a few electrons, such as hydrogen, fluorine, etc. It may, therefore, be worthwhile to look at the nature of the van der Waals radius from the point of view of the electron density distribution around an atom.

The electron density ψ^2 at distance r from the core of a hydrogenic one-electron atom is given by the well known relation

$$\psi^2 = C^2 \exp\left(-\frac{2\sqrt{2m_eI_0}}{\hbar}r\right) \qquad (1)$$

where m_e is the rest mass of the electron, I_0 the first ionization potential of the atom, and C is a normalization constant chosen such that $\int\psi^2 d\omega = 1$ when the integration is carried out over the whole of space (ω). The shape of the electron density distribution for typical one and multielectron atoms is shown on Fig. 1. As two atoms approach each other from $r = \infty$ their electron clouds interpenetrate more and more. The Pauli exclusion principle then causes a repulsion[7] of the two atoms in direct proportion[8] to the electron density in the region of interpenetration. One might define the van der Waals radius in terms of that distance r at which this repulsion just balances the attraction forces between the two atoms. Comparison of the abscissa of Fig. 1 with the empirically known van der Waals radii shows that this distance corresponds to a very low electron density—so low, in fact, that one cannot formulate a "critical" electron density and hope to calculate r_w from it because of the low degree of accuracy of the known calculating schemes in that part of the energy spectrum.

However, eq. 1 contains a parameter $h/\sqrt{m_eI_0}$, the de Broglie wave length λ_B of the outermost valence

(1) A. Bondi, *A.I.Ch.E. J.*, **8**, 610 (1962).

(2) A. Bondi and D. J. Simkin, *ibid.*, **6**, 191 (1960).

(3) Landolt-Bornstein, "Zahlenwerte and Funktionen I/4: Kristalle," Springer, Berlin, 1955.

(4) L. Pauling, "The Nature of the Chemical Bond," Cornell University Press, Ithaca, N. Y., 1942, p. 192.

(5) Since the volume increase of most solids between 0°K. and the melting point is about 10%, the van der Waals radii at 0°K. differ from those given here, probably by less than 3%.

(6) J. O. Hirschfelder and R. B. Bird, "Molecular Theory of Gases and Liquids," New York, N. Y., 1959.

(7) See J. A. A. Ketelaar, "Chemical Constitution," Elsevier Publishing Co., Amsterdam, 1953, p. 146 ff. for a more detailed discussion; a somewhat different treatment is presented by K. S. Pitzer, "Quantum Chemistry," Prentice-Hall, Inc., New York, N. Y., 1953, p. 201.

(8) The repulsion potential $E_{r2} \sim \exp(r/\alpha)$; hence, at the distances of the order r_w, considering eq. 1, $E_r \sim \psi^2$, the electron density in the region of interpenetration.

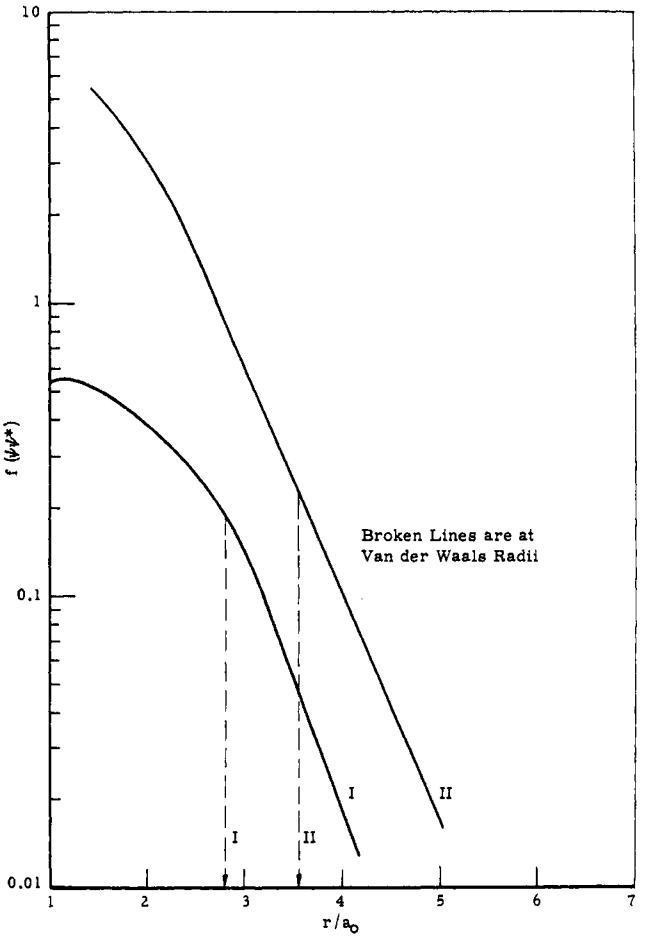


Figure 1. Electron density distribution near the atom "surface," for hydrogen (I) and argon (II) [D. R. and W. Hartree, *Proc. Roy. Soc.* (London), **A166**, 450 (1938)].

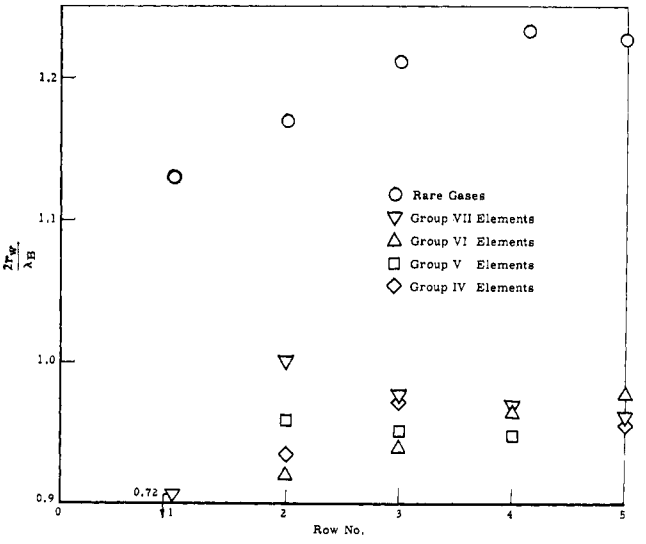


Figure 2. Relation of r_w (single bond value) to de Broglie wave length λ_B of outer valence electron for nonmetallic elements.

electron of an atom, which might be related to the van der Waals radius (r_w). A few years ago Morrison[9] suggested that perhaps r_w = (const.) λ_B. Our examination of this suggestion brought to light that this simple correlation holds surprisingly well. One finds that the constant is 0.61 for the rare gas atoms, and is—as one might expect—appreciably smaller for bound atoms, *ca.* 0.53 for the halogens and about 0.48 for the remainder of the nonmetallic elements. Deviations from this correlation (shown in Fig. 2) are in the direction of predicting too large a diameter for the lightest elements. As the valence electrons of a covalently bound atom are concentrated between it and its bound neighbor to a degree that depends on the nature of the bond, it is surprising that r_w retains enough individuality to be correlatable in terms of the ionization potential of the free atom.

Table I: Comparison of Correlation with "Observed" Mean van der Waals Radii (in Å.)

					H	He
r_b[a]					1.06	..
$\lambda_B/2$[b]					1.67	1.24
$\bar{r}_w$[c]					1.20	1.40
	B	C	N	O	F	Ne
r_b	1.65	1.53	1.46	1.42	1.40	..
$\lambda_B/2$	2.13	1.82	1.61	1.66	1.47	1.32
$\bar{r}_w$	..	1.70	1.55	1.52	1.47	1.54
	Al	Si	P	S	Cl	Ar
r_b	..	1.93	1.86	1.80	(1.75)[d]	..
$\lambda_B/2$	2.51	2.15	1.87	1.91	1.70	1.55
$\bar{r}_w$	..	2.10	1.80	1.80	1.75	1.88
	Ga	Ge	As	Se	Br	Kr
r_b	..	1.98	1.94	1.90	1.87	..
$\lambda_B/2$	2.51	2.19	1.96	1.97	1.79	1.64
$\bar{r}_w$	..	..	1.85	1.90	1.85	2.02
	In	Sn	Sb	Te	I	Xe
r_b	..	2.16	2.12	2.08	2.04	2.05
$\lambda_B/2$	2.55	2.27	2.09	2.05	1.90	1.76
$\bar{r}_w$	..	..	..	2.06	1.98	2.16

[a] $r_b \equiv b + 0.76$. [b] $\lambda_B \equiv 6.13 \times 10^{-8}/\sqrt{I}$. [c] Only the most frequently used values for single bonded forms of the elements are quoted here. [d] Reference point chosen for the system $r_b = b$ + constant.

Compatibility with Physical Properties

The range of numerical values of r_w for a given atom obtained from X-ray crystallographic contact distances is usually too wide for the direct application to a meaningful calculation of r_w. The "best" value of r_w must

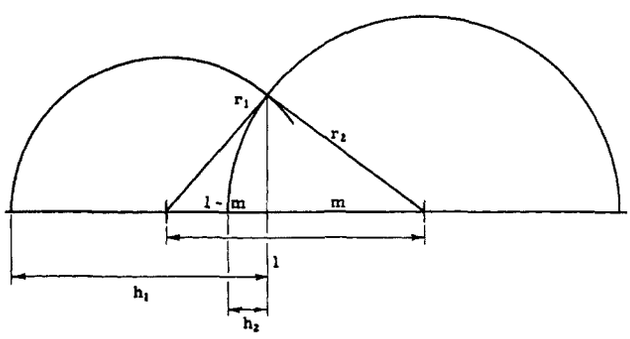
(9) J. D. Morrison, *Rev. Pure Appl. Chem.*, **5**, 46 (1955).

then be obtained by appeal to extraneous information. In the ideal case enough crystal structure and zero point density data are available to fix through calculation that value of r_w which yields the correct packing density ρ_0^* at 0°K. This has been done for the simple tetrahalides by Sackmann.[10]

In general, when a detailed analysis is either too laborious or not possible, the "plausibility" of the packing density calculated from the experimental zero point volume[11] and V_w will fix at least the upper limit of r_w. The empirical observation that for molecular crystals ρ_0^* always >0.6 may be used to fix a lower limit on V_w. The method used to estimate V_w from bond distance l and from r_w is shown on Fig. 3.

A lower limit is set for r_w by the requirement that the packing density at the critical temperature must be sufficiently high to permit the existence of a continuous three-dimensional network, *i.e.*, the number of nearest neighbors $Z \geq 3$.[12] A related criterium is the compatibility with the size of the equivalent sphere obtained by application of kinetic gas theory.[13] The somewhat surprising regularity that $V_w \approx N_A(\pi/6)\sigma^3$ which we observed whenever r_w met the other two criteria, is useful when neither ρ_0 nor ρ_c but only gas properties have been measured.

A particularly arbitrary criterium developed in the course of the present work is that the number of external degrees of freedom of rigid (nonlinear) polyatomic molecules calculated from liquid density and energy of vaporization ($E°$) data, as $0.391E°/RT \cdot (1.8) = 6 \pm 0.2$.[2] The application of this criterium requires the availability of good liquid density and vapor pressure or calorimetric heat of vaporization data and great faith in the form of the corresponding states principle adopted for this purpose.



r_1, r_2 = van der Waals radii
l = covalent bond distance
m = auxiliary parameter
h_1, h_2 = height of sphere segments

$$m = \frac{r_2^2 - r_1^2 + l^2}{2l}; \quad h_1 = r_1 + l - m; \quad h_2 = r_2 - m$$

$$V_1^1 = \pi h_1^2 (r_1 - \frac{h_1}{3}); \quad \Delta V_{2-1} = \pi h_2^2 (r_2 - \frac{h_2}{3}); \quad V_2 = \frac{4\pi}{3} r_2^3$$

Example:
van der Waals volume of diatomic molecule:

$$V_W = N_A [V_1^1 + V_2 - \Delta V_{2-1}] \text{ cm}^3/\text{mole}$$

where $N_A = 6.02 \times 10^{23}$ molecules/mole
and r's are given in Angstrom units

Volume of center atom = Total volume of atom
Surface area = $2\pi rh$

Figure 3. Method of calculation.

Experimental Data for Nonmetallic Elements

In this account the long form of the periodic table is followed from right to left.

The Halides. Fluorine. The data of Table II show that Pauling's suggestion to set the ionic radius (1.35 Å.) = r_w yields rather improbable values for ρ_0^* of well studied fluorine compounds. Crystallographic data for $NH_3 \cdot BF_3$,[14] SiF_4,[3,15] phosphonitrilic fluoride,[16] and Teflon[17,18] yield $r_w(F)$ = 1.50 Å., a value which, according to Table II, is compatible with ρ_0, ρ_c, and σ. The liquid state criterium led to the very similar value $r_w(F)$ = 1.47 Å. for perfluoroalkanes, aryl fluorides, and for secondary and tertiary alkane fluorides. However, for primary alkane fluorides $r_w(F)$ = 1.40 Å. was found more compatible with the data. In view of the exactly opposite trend of the C–F bond length, this is an unexpected result, for which neither conflicting nor confirmatory evidence could be found in X-ray diffraction data.

Chlorine, Bromine, and Iodine. The previously mentioned analysis of the crystallographic and density data of the tetrahalides of the group IV elements by Sackmann[10] yielded $r_w(Cl)$ = 1.76 Å., $r_w(Br)$ = 1.85 Å., $r_w(I)$ = 1.96 Å., in good agreement with many other X-ray diffraction data, shown in Table III. The data of Table IV indicate that the resulting values of V_w for various halogen compounds are more compatible with experimental density data than are the frequently quoted van der Waals radii based on Pauling's approximation $r_w = r_i$. The radii and volume increments derived from the liquid state criteria have been assembled in Table IV. They all show the same value for the primary alkanes previously noted for fluorine.

(10) H. Sackmann, *Z. physik. Chem.*, **208**, 235 (1958).
(11) W. Biltz, "Raumchemie der festen Stoffe," Leipzig, 1934.
(12) This point will be discussed in detail in a subsequent article.
(13) A. Bondi, *J. Phys. Chem.*, **58**, 929 (1954).
(14) J. L. Hoard, *et al.*, *Acta Cryst.*, **4**, 396 (1951).
(15) M. Atoji and W. N. Lipscomb, *ibid.*, **7**, 173 (1954).
(16) H. McGeachin and F. Tromans, *J. Chem. Soc.*, 4777 (1961).
(17) C. W. Bunn and E. R. Howells, *Nature*, **174**, 548 (1954).
(18) H. G. Killiam and E. Jenckel, *Z. Elektrochem.*, **63**, 308 (1959).

Table II: Compatibility of Proposed van der Waals Radii of Fluorine with Physical Properties

Substance	r_w(F), Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_0*[b,c]	ρ_0*[d]
F_2	1.35	10.5	0.88	0.42	...
	1.50	14.2	1.19	0.57	...
CF_4	1.35	16.5	0.51	0.56	0.161
	1.50	29.5	0.92	0.695	0.200
n-C_7F_{16}	1.35	105.8	..	...	0.160
	1.50	127.7	..	...	0.193

[a] See ref. 6. [b] See ref. 11. [c] G. A. Miller and R. B. Bernstein, *J. Phys. Chem.*, **63**, 710 (1959). [d] K. A. Kobe and R. E. Lynn, *Chem. Rev.*, **52**, 121 (1953), and Miller and Bernstein, preceding reference.

Table III: Recent Intermolecular Contact Radii r' for Chlorine, Bromine, and Iodine in Molecular Crystals

Compound	r_w'(X), Å.
λ-$C_6H_6Cl_6$ (Gammexane)	(Cl) 1.70[a]
Perchlorofulvalene	(Cl) 1.74[b]
Tetrachloro-p-benzoquinone	(Cl) 1.78[c]
1,3,5-Trichlorobenzene (90°K.), 293°K.	(Cl) (1.77), 1.82[d]
1,4-Dibromocyclo[3.2.2]azine	(Br) 1.80[e]
p-Dibromobenzene	(Br) 1.88[f]
1,3,5-Tribromobenzene	(Br) 1.88[d]
Iodoform, 0°K.	1.99[g]
Phosphorus thioiodide	1.95[h]
p-Diiodobenzene	2.07[i]

[a] G. W. Van Vlater, *et al.*, *Acta Cryst.*, **3**, 139 (1950). [b] See ref. 21. [c] S. Chu, *et al.*, *Acta Cryst.*, **15**, 661 (1962). [d] H. J. Milledge and L. M. Pant, *ibid.*, **13**, 285 (1960). [e] A. Hanson, *ibid.*, **14**, 124 (1961). [f] U. Croatto, *et al.*, *ibid.*, **5**, 825 (1952). [g] See ref. 30. [h] D. A. Wright and B. R. Penfold, *Acta Cryst.*, **12**, 455 (1959). [i] S. B. Hendricks, *et al.*, *J. Chem. Phys.*, **1**, 549 (1933).

Table IV: Compatibility of Proposed van der Waals Radii of Cl, Br, and I with Physical Properties

Substance	r_w(X), Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_0*[b]	ρ_0*[c]
Cl_2	1.75	24.1	1.09	0.74	0.195
Br_2	1.85	28.9	1.16	0.75	0.200
	2.00	35.6	1.42	0.93	0.240
I_2	1.96	37.6	0.96	0.76	...
	2.16	46.0	1.17	0.93	...
CCl_4	1.75	51.4[d]	0.8	0.69	0.186
CBr_4	1.85	60.8[d]	0.84	0.69	...
CI_4	1.97	74.9[d]	..	0.68	...

[a] See ref. 6 and R. A. Svela, NASA Technical Report R-132 (1962). [b] See ref. 11. [c] See ref. *d* in Table II. [d] No allowance has been made for possible volume reduction due to overlap of the halogen atoms or for their probable pear shape. Such correction would reduce V_w and the other reduced densities, but would leave the constancy in the series unchanged.

Table V: Effective van der Waals Radii of Bound Halogen Atoms from Liquid State p-v-t Property Calculations

X	Attachment	r_w(X), Å.
F	Primary alkyl	1.40
	sec, *tert*, alkyl, perfluous alkane, phenyl	1.47
Cl	Primary alkyl	1.73
	Vinyl	1.75
	sec, *tert*, alkyl, polychloroalkyl, phenyl	1.77
Br	Primary alkyl	1.84
	tert, polybromoalkyl	1.85
	Phenyl	1.92
I	Primary alkyl	2.01
	tert, polyiodoalkyl, phenyl	2.06

Oxygen, Sulfur, Selenium, and Tellurium. As the diameter of singly bonded oxygen and sulfur is smaller than that of methylene or other functional groups to which they might be attached, one finds only few good contact distance data for them. Moreover, in the present context all contacts involving hydrogen bonds must be excluded. As a consequence of being too deeply "buried" within the molecule for frequent (or any) collision with neighboring molecules, the "effective volume" occupied by an ether oxygen atom differs from molecule to molecule. The upper limit on $r_w(>O)$ set by the few available X-ray data (Table VI), 1.52 Å., corresponds to $V_w(>O)$ = 5.5 cm.³/mole. Only in ethylene oxide is the ether oxygen sufficiently exposed to yield this value of $V_w(>O)$. In alkyl ethers, especially polyethers, $V_w(>O)$ = 3.7 cm.³/mole and in polyphenyl ethers it is 3.2 cm.³/mole. These values were obtained by fitting polyether data to the generalized density correlation for liquids.

For single-bonded sulfur, X-ray diffraction data point to $r_w(>S)$ = 1.83 Å., while 1.80 Å. is compatible with various physical properties (Table VII) and with liquid phase density. For single-bonded selenium, a single set of data[19] yields $r_w(>Se)$ = 1.87 Å. and for tellurium V_w (>Te) = 2.06 Å.

The geometry of double-bonded oxygen and sulfur is probably quite anisometric. In the direction parallel

(19) R. E. Mush, *Acta Cryst.*, **5**, 458 (1952).

Table VI: Intermolecular Contact Radii for Oxygen[a]

Compound	$r_w'(O)$
Single-bonded (ether) oxygen	
Calcium peroxide	1.46[g]
1,3,6-Trioxane	~1.5[b]
Bis(1,3-dioxacyclopentyl)	1.47[b]
Hydroxy L-proline	1.52[c]
Double-bonded (carbonyl or nitro) oxygen	
Tetrachloro-*p*-benzoquinone, parallel arrangement	1.35[d]
Tetrachloro-*p*-benzoquinone, vertical arrangement	1.63[d]
Butyne-2, iron octacarbonyl, parallel arrangement	1.47[e]
Acrylonitrile iron tetracarbonyl	1.53[a,f]

[a] Only O...O contact are quoted here. The recommended van der Waals radii also take into account the intermolecular contact distances between oxygen and other atoms. [b] See ref. 3. [c] J. Donohue and K. N. Trueblood, *Acta Cryst.*, **5**, 419 (1952). [d] See ref. *c* in Table III. [e] A. Hock and O. Mills, *Acta Cryst.*, **14**, 139 (1961). [f] A. Luxmoore and M. Truter, *ibid.*, **15**, 1117 (1962). [g] N. G. Vannenberg, *Progr. Inorg. Chem.*, **4**, 125 (1962).

Table VII: Compatibility of van der Waals Radii of Sulfur and Selenium with Physical Properties

Substance	$\bar{r}_w$, Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_0*[b]	ρ_c*[c]
H_2S	1.80 (S<)	18.0	1.21	0.675	0.186
S_8	1.75 (–S–)	92.0	..	0.766	...
CS_2	1.75 (S)	31.2	1.09	0.69	0.184
H_2Se	1.90 (Se<)	20.9	..	0.68	...

[a] See ref. 6. [b] See ref. 11. [c] See ref. *c* in Table II.

to the double bond, one finds $r_w(O) = 1.40$ Å. (about 1.39 Å. is the cut-off distance of the neutron diffraction pattern of liquid oxygen,[20] and 1.43 ± 0.03 Å. for similar dimensions in various carbonyl compounds. Normal to the double bond axis, one finds dimensions of the order of 1.6 to 1.7 Å., perhaps characteristic for the diameter of a π-electron cloud. Owing to the uncertainty of the geometry of the π-electron cloud, the arbitrary compromise $r_w(=O) = 1.50$ Å. was adopted, which is compatible with the density data, as shown in Table VIII, and meets the requirements of the liquid state correlation.

The contact distance involving double-bonded sulfur, as in rhodanin[21] [$r_w(S) = 1.74$ Å.] is smaller than in single-bonded sulfur. Similarly, the contact distance between sulfur atoms in polysulfides and in crystalline sulfur[22] is comparatively short ($r_w = 1.76 \pm 0.04$ Å.), probably connected with the pronounced double-bond character of the S–S bond.

Table VIII: Compatibility of van der Waals Radius of Double-Bonded Oxygen with Physical Properties

Substance	$\bar{r}_w(O)$, Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_c*[b]	ρ_c*[c]
O_2	1.50	13.0	1.03	0.60	0.167
C=O	1.50	16.2	1.01	0.60	0.174
CO_2	1.50	19.7	0.98	0.76	0.21

[a] See ref. 6. [b] See ref. 11. [c] See ref. *c* in Table II.

Nitrogen, Phosphorus, and Arsenic. The contact distances in a wide variety of nitrogen compounds, assembled in Table IX, suggest that for single- and double-bonded nitrogen $r_w(N) = 1.55 \pm 0.04$ Å. Triple-bonded nitrogen, as in N_2 and in cyano groups, appears to be quite anisometric with $(r_w) = 1.40$ Å. parallel to the bond and up to 1.7 Å. in the direction normal to bonds. Again the "effective" value was somewhat arbitrarily set at $r_w(N) = 1.60$ Å. The data of Table X show that the adopted radii for nitrogen are compatible with the density data. They also yield the plausible packing density $\rho_s^* = 0.706$ for the carefully investigated (tightly packed) crystal of tetracyanoethylene.[23] Smaller values for $r_w(N)$ would not have given a packing density in keeping with X-ray observations. Further confirmation of the proposed radii comes from the liquid state correlation.

Only very few contact distance data involving phosphorus are available.[16,24] Their average, $r_w(P) = 1.80$ Å., is of the order expected from the ionization potential and from the covalent bond radius of phosphorus. Only one set of crystallographic data (bromodiphenylarsine) could be found from which $r_w(As)$ could be estimated as ≈ 1.85 Å. This value agrees with the physical property data (Table XI).

Carbon and Hydrogen. The available data for the van der Waals radii of carbon in aliphatic and aromatic compounds have been assembled in Table XII. Olefinic carbon is not found in the list because only the so-far unexamined cumulenes, such as allene, could yield the desired contact distances, not being spaced apart by the always larger methyl or functional groups. The data obtained on the various acids and amides on the list could be considered as measures of the width of a double-bonded carbon atom, because the reported spacings are those characteristic of the

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(22) J. Donohue in "Organic Sulfur Compounds," N. Kharasch, Ed., New York, N. Y., 1961, p. 1.

(23) D. A. Bekoe and K. N. Trueblood, *Z. Krist.*, **113**, 1 (1960).

(24) E. Kessler and A. Vos, *Acta Cryst.*, **12**, 323 (1959).

Table IX: Intermolecular Contact Radii for Nitrogen[a]

Compound	$r_w'(N)$, Å.
Single-bonded	
Xanthazol	1.48[b]
$S_4N_4H_4$	1.58[c]
Double-bonded	
4-Methylimidazole	1.50[d]
Glyoxime	1.50[e]
Guanine	1.54[f]
Pyrimidine	1.56[g]
Adenine hydrochloride	1.58[f]
Diazoaminobenzene copper	1.68[h]
Tetrazole	1.63[i,j]
Triple-bonded	
Tetracyanoethylene (N...N contact, normal to bond)	1.70[k]
Tetracyanoethylene (N...C contact, parallel with bond)	1.39[k]

[a] Only N...N contacts are quoted here, except where noted otherwise. The recommended van der Waals radii also take into account the intermolecular contact distances between nitrogen and other atoms. [b] W. Nowacki and H. Burki, *Z. Krist.*, **106**, 339 (1955). [c] R. L. Sass and J. Donohue, *Acta Cryst.*, **11**, 497 (1958). [d] H. Zimmerman, *Ann.*, **612**, 193 (1958). [e] W. Hamilton, *Acta Cryst.*, **14**, 95 (1961). [f] J. M. Broomhead, *ibid.*, **1**, 324 (1948); **4**, 92 (1951). [g] N. E. White and C. J. B. Clews, *ibid.*, **9**, 586 (1956). [h] J. H. Bryden, *ibid.*, **8**, 211 (1955). [i] R. N. Brown, *ibid.*, **9**, 163 (1956). [j] Y. D. Kondenshev, *Soviet Phys. Cryst.*, **6**, 413 (1962). [k] See ref. 23.

Table X: Compatibility of van der Waals Radii of Nitrogen with Physical Properties of Its Compounds

Substance	$\bar{r}_w(N)$, Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_0^{*b}	ρ_c^{*c}
N≡N	1.60	15.8	1.01	0.59[d]	0.176
$(C{\equiv}N)_2$	1.60	29.4	1.10	..	...
N=O	1.55	13.9	1.07	0.73	...
N=N=O	1.55	18.9	1.03	0.70	0.196
NH_3	1.55	13.8	..	0.70	0.190

[a] See ref. 6. [b] See ref. 11. [c] See ref. *c* in Table II. [d] L. H. Bolz, *et al.*, *Acta Cryst.*, **12**, 247 (1959).

Table XI: Compatibility of van der Waals Radii of Phosphorus and Arsenic with Physical Properties of Their Compounds

Substance	$r_w(X)$, Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_0^{*b}	ρ_c^{*c}
P_4	1.80	41.6	(0.81)	0.676	...
PH_3	1.80	19.7	0.986	0.55	0.174
AsH_3	1.85	21.3	1.00	0.56	...

[a] See ref. 6. [b] See ref. 11. [c] See ref. *c* in Table II.

carboxylic carbonyl group. The single datum for the radius of the acetylenic triple bond is supported by the volume data in Table XIII.

Table XII: Observed Contact Radii of Carbon and Hydrogen

Aliphatic compounds	$r_w'(C)$, Å.	$r_w(H)$, Å.	Ref.
1,3-Dimethyliminotetrazole·HCl	1.66	..	*a*
Uracil	1.67	..	*b*
Sodium tropolonate	1.67	..	*c*
Octatriyne	1.69[d]	..	*e*
Succinamide	1.69; 1.70	..	*f*
Isocyanic acid	1.70	..	*g*
Succinamic acid	1.70	..	*h*
n-Triacontane at 0°K.	..	1.17	
n-Hexatriacontane	..	1.18	*i*
Triethylenediaminenickel(II) nitrate	..	1.18	*j*
Potassium methylene disulfonate	..	1.17	*k*
Acetylenic triple bonds			
Carbides of Ca, Ba, and Sr	1.78	..	*l*
Aromatic ring systems			
Coronene	1.72	..	*m*
Ovalene, perylene	1.75	..	*n*
Anthraquinone	1.77	..	*o*
Naphthalene	1.77	1.01	*p*
Anthracene	1.77	1.01	*q*
Graphite	1.78	..	*r*
Dibenzocoronene	1.80	..	*s*
Benzene	..	1.01; 1.05	*t*
Pyridine·BF_3	..	1.0	*u*

[a] See ref. *h* in Table IX. [b] G. S. Parry, *Acta Cryst.*, **7**, 313 (1954). [c] R. Shiono, *ibid.*, **14**, 42 (1961). [d] Head-on approach of methyl groups. [e] See ref. 3. [f] R. A. Pasternak, *Acta Cryst.*, **6**, 808 (1953); **9**, 334 (1946). [g] See ref. 3. [h] R. A. Pasternak, *Acta Cryst.*, **7**, 225 (1954). [i] P. W. Teare, *ibid.*, **12**, 294 (1959). [j] L. Swink and M. Atoji, *ibid.*, **13**, 639 (1960). [k] M. Truter, *J. Chem. Soc.*, 3393 (1962). [l] M. Atoji and R. C. Medrud, *J. Chem. Phys.*, **31**, 332 (1959). [m] J. M. Robertson, *et al.*, *J. Chem. Soc.*, 925 (1956). [n] See ref. 28. [o] S. N. Sen, *Indian J. Phys.*, **22**, 347 (1948). [p] D. W. T. Cruikshank, *Acta Cryst.*, **10**, 504 (1957). [q] D. W. T. Cruikshank, *ibid.*, **9**, 915 (1956). [r] J. A. Ibers, private communication. [s] See ref. 29. [t] E. G. Cox, *Rev. Mod. Phys.*, **30**, 159 (1958). [u] S. V. Svenkova, *Kristallografiya*, **2**, 408 (1957).

The head-on approach of the methyl groups on octatriyne provides the most direct measure of the dimensions of an aliphatic carbon atom. However, adoption of $r_w(C) = 1.69$ Å. would have inspired a false sense of precision regarding this central factor in the entire scheme of volume calculations. Hence $r_w(C_{aliph}) = 1.70$ Å. is, in agreement with Briegleb[25] and others,[26] the preferred choice at present.

Table XIII: Compatibility of van der Waals Radii of Carbon and Hydrogen with Physical Properties of Simple Hydrocarbons

Substance	r_w(C), Å.	r_w(H), Å.	V_w, cm.³/mole	$\frac{V_w}{N_A(\pi/6)\sigma^3}$ [a]	ρ_0*[b]	ρ_0*[c]
Methane	1.70	1.20	17.1	1.00	0.67	0.173
Ethane	1.70	1.20	27.3	1.00	0.684	0.184
Ethylene	1.70	1.20	23.9	1.00	0.67	0.193
Acetylene	1.78	1.20	23.1	0.98	..	0.204
Benzene	1.77	1.00	48.4	1.04	0.70	0.186

[a] See ref. 6. [b] See ref. 11. [c] See ref. *c* in Table II.

The van der Waals radius of aromatic carbon atoms is one-half of the contact distance between nonbonded carbon atoms in adjacent ring planes and is distinctly different from the often quoted interplanar distances. A few shorter distances (as low as 3.20 Å.) between ring carbons in neighboring rings disposed at steep angles have been reported[27-29] but have not been given any weight in the final estimate of $r_w(C_w)$. The value $r_w(C_w)$ = 1.77 Å. has been adopted as being authenticated by the most careful measurements on the list. The supposition is then, that the benzene ring has a slightly bumpy top (like the top of an old Arab house). The smooth π-electron model of Briegleb does not seem entirely compatible with the observed X-ray diffraction patterns of condensed ring systems. However, the difference in volume between the two models is small.

Neither model predicts the density of graphite correctly. Our model yields a density that is about 5% too high and that of the Briegleb model is about 5% too low. The model predicting too high a density is probably more realistic because it means that there is still some space for thermal motion between the graphite planes. From Bridgman's experiments we know that about 5% of graphite volume is squeezed out by compression to about 10,000 atm.

The contact distances of hydrogen are difficult to ascertain. The best authenticated contacts, by Kitaigorodskii's measurements on *n*-paraffins extrapolated to 0°K.[26,30] and also observed in several other crystals lead to $r_w(H_{aliph})$ = 1.17 Å. This is so close to the commonly accepted value of $r_w(H_{aliph})$ = 1.20 Å. that the latter value was retained for the present work. This choice of aliphatic radii is compatible with the shortest observed contact distance of methyl groups, 3.57 Å., in gear-like packing of aliphatic compounds, as well as with densities and gas properties of small hydrocarbon molecules, as shown by the data of Table XIII. The observed hydrogen–hydrogen contact distances between aromatic rings in the ring plans of several very careful measurements lead uniformly to $r_w(H_w)$ = 1.0 Å. The choice of aromatic dimensions is also compatible with density data and gas properties, as shown in Table XIII.

Silicon, Germanium, and Tin. No crystallographic contact distances were found for these elements in nonmetallic and nonionic crystals. Their van der Waals radii must, therefore, be estimated from correlations, either with covalent bond radii or with ionization potentials. The density of pure silicon and of SiH_4 (liq) as well as ρ_0 [ρ_0*(SiH_4) = ρ_0*(CH_4) = 0.67] are compatible with a van der Waals radius of 2.1 Å.

Metals (in Nonmetallic Compounds). The van der Waals radii of metals in metal organic compounds are very difficult to obtain, because in most crystalline compounds observed to date the metal atom is hidden from nonbonded contacts. Two recent measurements yield rather short radii; in diazoaminobenzene–copper(I)[31] a C. . .Cu contact yields r_w(Cu) = 1.43 Å., and in methoxy carbonyl mercuric chloride[32] the (nonbonded) Hg. . . .Cl and the Hg. . . .O contacts yield r_w(Hg) = 1.5 Å. The latter is only slightly smaller than the value for r_0 estimated from gas viscosity measurements of mercury vapor, namely $r_0/2$ = 1.65 ± 0.03 Å. and that estimated from the critical volume (by a method to be discussed in another paper), $r_0/2$ = 1.55 Å. Other radii derived from critical volumes are assembled in Table XI.

Spectroscopic evidence for the cessation of d-shell, *i.e.*, metallic, interaction in dialkyl glyoxime complexes, when steric hindrance forced the metal atoms apart, indicates that for nickel and palladium r_w = 1.63 Å. and for platinum r_w = 1.7 to 1.8 Å.[33] All of the aforementioned metal radii are substantially smaller than predicted from ionization potentials and somewhat larger, but not uniformly larger, than their covalent bond radii. It is not certain, therefore, how best to estimate the van der Waals radii of metals in metal organic compounds.

(25) G. Briegleb, *Fortschr. Chem. Forsch.*, **1**, 642 (1950).
(26) A. I. Kitaigorodskii, "Organic Chemical Crystallography," Consultants Bureau, New York, N. Y., 1961.
(27) A. O. McIntosh, *et al.*, *J. Chem. Soc.*, 1661 (1954).
(28) J. M. Robertson, *et al.*, *ibid.*, 925 (1956).
(29) J. M. Robertson and J. Trotter, *ibid.*, 1115 (1961).
(30) A. I. Kitaigorodskii and Y. V. Muinkh, *Soviet Phys. Dokl.*, **3**, 707 (1958).
(31) I. Brown and J. Dimitz, *Acta Cryst.*, **14**, 480 (1961).
(32) T. Mak and J. Trotter, *J. Chem. Soc.*, 3243 (1962).
(33) C. R. Banks and D. W. Barnum, *J. Am. Chem. Soc.*, **80**, 3579, 4767 (1958).

Table XIV: Radii of Metal Atoms in Nonbonded State, from Critical Volumes[a] (Except where Noted Otherwise)[b] in Å.

Li 1.82						
Na 2.27	Mg 1.73					
K 2.75		Ni 1.63[c]	Cu 1.4[d]	Zn 1.39	Ga 1.87	
		Pd 1.63[c]	Ag 1.72	Cd 1.58	In 1.93	Sn 2.17
		Pt 1.75[c] 1.72[d]	Au 1.66	Hg 1.55	Tl 1.96	Pb 2.02
	U 1.86					

[a] P. J. McGonigal, *J. Phys. Chem.*, **66**, 1686 (1962). [b] $r_w = {}^1/_2((0.66/\pi N_A)V_c)^{1/3} = {}^1/_2(0.70 \times 10^{-24}V_c)^{1/3}$. [c] From ultraviolet spectroscopy[33] of glyoxime complexes. [d] From X-ray diffraction data: Cu (ref. 31) and Pt [see R. E. Rundle and J. H. Sturdivant, *J. Am. Chem. Soc.*, **69**, 1561 (1947)].

Table XV: Group Contributions to the van der Waals Volume and Surface Area of Hydrocarbons[a]

Groups	V_w, cm.³/mole	A_w, cm.²/mole × 10⁹
Alkane		
—C—	3.33	0
—CH	6.78	0.57
>CH₂	10.23	1.35
—CH₃	13.67	2.12
CH₄	17.12	2.90
n-Paraffins	$6.88 + 10.23\,N_c$	$1.54 + 1.35\,N_c$
Olefinic		
=C=	6.96	..
>C=C<	10.02	0.61
=CH	8.47	1.08
=CH₂	11.94	1.86
>C=CH₂	16.95	2.17
>C=CH	13.49	1.39
Acetylenic		
—C≡	8.05	0.98
≡C—H	11.55	1.74
≡C— (in diacetylene)	7.82	..
Aromatic		
⋗C— (condensation)	4.74	0.21
⋗C— (alkyl)	5.54	0.30
⋗C—H	8.06	1.00
Benzene	48.36	6.01
Phenyl	45.84	5.33
Naphthalene	73.97	8.44
Naphthyl	71.45	7.76

[a] See corrections for crowding and vicinal effects on Table XVII.

Summary Tables of Group Increments

For the convenience of the user, a number of frequently occurring group increments have been calculated and assembled in Tables XV and XVI. It should be noted that the group increments in Table XVI are correct only for attachment to carbon atoms. The example of ether oxygen illustrates the seriousness of the error due to neglect of this restriction: in certain alkanes $V_w(-O-) = 5.20$ cm.³/mole and in siloxanes it is 2.7 cm.³/mole.

Corrections for Intramolecular Crowding and for Association Bonding. The simple model for the calculation of the van der Waals volume V_w of molecules shown in Fig. 3 is not always adequate.

1. Care must be taken not to cut larger sphere segments off the central atom than it has volume to provide.

2. Neighboring atoms around a three of four valent central atom would interpenetrate if the sphericities of their shells were preserved to the root. Actually, they do not even squeeze each other very hard, as the bound atoms are probably pear-shaped due to the tighter binding of valence electrons halfway between the two radii. Because of insufficient data on dimensions in this region, we ignored the problem in the calculation of V_w.

3. More important is the crowding effect in saturated ring structures such as cyclohexane. Typical crowding decrements calculated for these are given in Table XVII.

4. Strong association causes in some cases sufficient deformation of the participating atoms that it has to be taken into consideration in the selection of contact distances and in the calculation of V_w.

The best known example of such deformation is that of hydrogen-bonded systems where interatomic distances at times decrease by 0.8 Å. below those expected from the usual contact distance data. An example for sharp decreases of d_w without hydrogen bonding is the short intralayer distance between nonbonded halide atoms in the crystals of Cl_2, Br_2, and I_2.[34-36] Guided by the principle that one should adopt the smallest observed intermolecular distances as source of r_w data, one might have been tempted to use these, namely 1.66, 1.67, and 1.77 Å. instead of 1.75,

Table XVI: Functional Group Contributions to van der Waals Volumes and Surface Areas, All Attached to Carbon Atoms[a]

Groups	V_w, cm.³/mole	A_w, cm.²/mole × 10⁹
—O— (c.e.)[b]	5.20	0.74
—O— (a.e.)[b]	3.7	0.6
—O— (ph.e.)[b]	3.2	0.54
—OH[c]	8.04	1.46
>C=O[d]	11.70	1.60
—S—	10.8	..
—SH	14.8	..
Nitrogen and phosphorus		
NH_2	10.54	1.74
>NH	8.08	0.99
—N—	4.33	0.23
—C≡N	14.70	2.19
$—NO_2$	16.8	2.55
>—P	10.44	..
Fluorine		
—F (pr)	5.72	1.10
—F (s,t)	6.20	1.18
—F (p)	6.00	1.15
—F (ph)	5.80	1.10
Chlorine		
—Cl (pr)[e]	11.62	1.80
—Cl (s,t,p)	12.24	1.82
—Cl (v)	11.65	1.80
—Cl (ph)	12.0	1.81
Bromine and iodine		
—Br (pr)	14.40	2.08
—Br (s,t,p)	14.60	2.09
—Be (pr)	15.12	2.13
—I (pr)	19.18	2.48
—I (s,t,p)	20.35	2.54
—I (ph)	19.64	2.51

[a] See ref. *f* in Table IX. [b] c.e., heterocycloaliphatic ethers; a.e., (poly)alkane ethers; ph.e., (poly)phenyl ethers. [c] Correction for hydrogen bonding, *e.g.*, O...O (2.78 Å.): subtract 1.05 cm.³ mole per hydrogen bond. [d] Corrections to carboxylates and amides in Table XVII. [e] pr, attached to alkane in primary position; s,t, attached to alkane in secondary or tertiary position; p, per- or polyhalide of alkane; v, attached to vinyl group; ph, attached to phenyl ring. [f] Adjusted to yield the requisite number of external degrees of freedom in rigid molecules in the liquid state. The second-order refinements on the halides are due to this constraint.

1.85, and 1.97 Å. as van der Waals radii for Cl, Br, and I, respectively. The very much higher heat of sublimation increment per halide atom observed for these X_2 crystals than for other halogen compounds was an additional warning signal against the adoption of these anomalously small radii.

Table XVII: Decrements in van der Waals Volume and Surface Area due to Intramolecular Crowding and Related Effects

Description	δV_w, cm.³/mole	δA_w, cm.²/mole × 10⁹
Decrement per cyclohexyl and per cyclopentyl ring, free and in transcondensed cyclic naphthenes	1.14	0.57
Decrement per ring in *cis*-condensed cyclic naphthenes[a]	2.50	1.2
Decrement per methylene ring condensed to benzene[b]	1.66	0.7
Decrement per dioxane ring	1.70	0.7
Decrement per single bond between conjugated double bonds	0.25	..
Decrement per single bond adjacent to carboxyl or amide group	0.22	..

[a] This appears to be the correct decrement to use with the isomer mixtures produced by the hydrogenation of these ring and higher condensed cyclic aromatics. [b] Or other aromatic ring system; examples: tetralin and indane.

While the short distances between halogen molecules are restricted to these molecules and to the crystalline state, X-ray evidence points to the fact that the shorter distance between hydrogen-bonded atoms prevails in the liquid as well as in the crystalline states for molecules containing hydrogen-bonding groups.[37] The corresponding correction of V_w therefore cannot be neglected. Figure 4 gives the magnitude of these corrections for several typical hydrogen bonds.

Comparison with Other Work.[9] A very similar effort to the present one was undertaken by Kitaigorodskii[26] who, among X-ray crystallographers, has been most consistently concerned with the establishment of reliable intermolecular van der Waals radii. The radii proposed by him have been given considerable weight in arriving at recommended values. It is not surprising, therefore, that many of his V_w increments, shown in Table XVIII, are similar to those of this work.

Two other authors, Edward[38] and Luzkii,[39] have proposed the calculation of V_w from l and r_w.[40] Both made the simplifying assumption that the volume increment of an atom is a function only of its own

(34) R. L. Collin, *Acta Cryst.*, **5**, 431 (1952).
(35) B. Vonnegut and B. E. Warren, *J. Am. Chem. Soc.*, **58**, 2459 (1936).
(36) J. Harris, *ibid.*, **50**, 1583 (1928).
(37) G. G. Harvey, *J. Chem. Phys.*, **6**, 111 (1938); **7**, 878 (1939).
(38) J. T. Edward, *Chem. Ind.* (London), 774 (1956).
(39) A. E. Luzkii, *Dokl. Akad. Nauk SSSR*, **44**, 513 (1954); *Russ. J. Phys. Chem.*, **28**, 204 (1954).

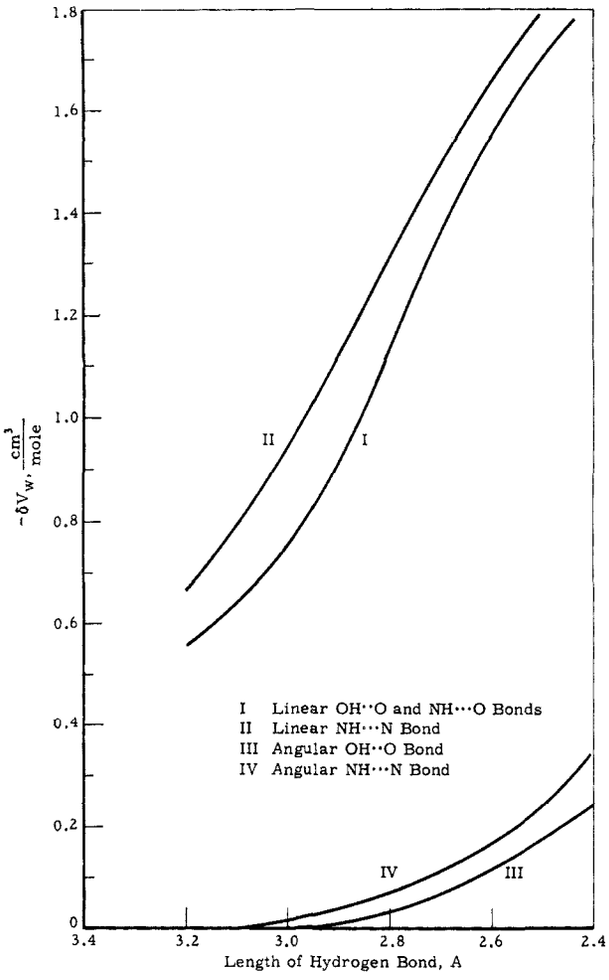

Figure 4. van der Waals volume correction (δV_w) for hydrogen bond formation.

Table XVIII: Comparison of Several Group Increments of V_w Computed by Different Authors with the Present Work, in Cm.^{3}/Mole

Group	This work	Edward[a]	Kitaigorodskii[b]	Luzkii[c]
$>CH_2$	10.23	9.90	10.0	8.84
$-CH_3$	13.67	13.04	13.32	11.61
$-\overset{\vert}{\underset{\vert}{C}}-$ (aliph.)	3.33	3.62	3.31	3.30
$\geqslant C-$ arom. cond.	4.74	...	4.70	...
$\geqslant C-$ arom. alkyl	5.54	...	5.06	4.9
$\geqslant C-H$ arom.	8.06	...	8.37	7.70
$\equiv C-$	8.05	8.40	...	...
$-F$	5.72 (aliph.)	5.17	...	...
$-Cl$	11.62 (aliph.)	12.9	12.78	12.8
$-Br$	14.40 (aliph.)	16.7	16.6	15.5
$>O$	5.20	4.51	...	5.77
$=O$	6.70	7.05	...	6.50
$-\overset{\vert}{\underset{\vert}{N}}-$	4.33	4.10	...	4.85
$>S$	10.8	12.1	...	12.05
$-NO_2$	11.8	...	13.87	...

[a] See ref. 38. [b] See ref. 26. [c] See ref. 39.

Table XIX: Comparison of V_w with the Molecular Volume Calculated for End Group Data by Hubert and Schultze[a]

Substance	V_w, cm.^{3}/mole	V(H + S),[b] cm.^{3}/mole
Ethylene	23.9	25.5
Benzene	48.4	49.2

[a] See ref. 41. [b] Hubert and Schultze's data have been corrected by the factor 10.23/11.1 in order to be consistent with the CH_2 model of this report.

covalent bond and van der Waals radii and is unaffected by the dimensions of the atom to which it is attached. The errors that can be caused by following this argument have been pointed out earlier in this paper.

The differences between the V_w increments of these authors and those of the present work, on Table XVIII, are largely due to the use of the old (Pauling) values for r_w. Edward's high increments for all multiple bond systems is due to his use of Briegleb's model for the π-electron cloud.

Rather interesting is the agreement between the van der Waals volume of various hydrocarbon end groups derived by Hubert and Schultze[41] from the density relations of homologous series and V_w calculated here, as shown in Table XIX. The adjustment for the more accurate volume of the CH_2 group (10.23 cm.^{3}) instead of that used by Hubert and Schultze (11.1 cm.^{3}) based on the over-all radius of the CH_2 group of 2.0 Å. is quite small.

It cannot be overemphasized that the van der Waals radii of this paper have been selected for the calculation of volumes. They may not always be suitable for the calculation of contact distances in crystals.

Acknowledgment. The author is made indebted to Mr. Janis Lielmesz for diligent search of the crystallographic literature and for his help with some of the calculations, and to Mr. A. E. Smith for valuable discussions.

(40) A limited collection of van der Waals radii for use in inorganic systems, but without supporting evidence for their choice, has just been presented by W. Klemm and E. Busmann, *Z. anorg. allgem. Chem.*, **319**, 297 (1963).

(41) H. J. Hubert and G. K. Schultze, *Z. Elektrochem.*, **61**, 126 (1957).